SECTION 23 07 13.01

DUCTWORK INSULATION, SCHEDULE

Rev #	Description of Change	Author	Package	Date
23 07 13.01				
A	Initial issue; Issue for Final Review	Ankush Shelar	PS-05	05 Sep 2025
0	Issue for Construction	Ankush Shelar	PS-05	26 Sep 2025

SECTION 23 07 13.01 DUCTWORK INSULATION SCHEDULE, CHANGE SCHEDULE	
Schedule Number	**Version**

SECTION 23 07 13.01

DUCTWORK INSULATION SCHEDULE 1

FAB Support Buildings and Site Buildings (Note 2)									
Plenum or Ductwork Type	**Duct Type**	**Climate Zone**	**Plenum or Ductwork Location**		**Insulation**			**Jacketing Note 1**	**Notes**
			Indoor	**Outdoor**	**Type(s) Note 1**	**R-value (min.) $m^2.K/W$**	**Thickness (basis of design) mm**		
Outside Air Plenums in an Unconditioned Space	Rectangular	0B	X		NR				
Outside Air Plenums in Conditioned Space	Rectangular	0B	X		E	0.26	9	Included	
Relief and Exhaust Plenums	Rectangular	0B	X		NR				
Louver Blank Off Panels	Rectangular	0B	X		NR				
Downstream (Supply) and Upstream (Return) of Air Handling Units	Rectangular	0B	X		L2	0.26	9	N/A	4
Outside Air Ductwork Exposed in Conditioned Space	Rectangular	0B	X		E	0.26	9	Included	
	Round	0B	X		E	0.26	9	Included	
	Oval	0B	X		E	0.26	9	Included	
Outside Air Ductwork Exposed in an Unconditioned Space	Rectangular	0B	X		NR				
	Round	0B	X		NR				
	Oval	0B	X		NR				
Makeup Air Ductwork Exposed - Outdoors	Rectangular	0B		X	G	0.91	32	Included	5
	Round	0B		X	G	0.91	32	Included	5

FAB Support Buildings and Site Buildings (Note 2)									
Plenum or Ductwork Type	Duct Type	Climate Zone	Plenum or Ductwork Location		Insulation			Jacketing Note 1	Notes
			Indoor	Outdoor	Type(s) Note 1	R-value (min.) $m^2.K/W$	Thickness (basis of design) mm		
Makeup Air Ductwork Exposed in Conditioned Space	Rectangular	0B	X		E	0.26	9	Included	
	Round	0B	X		E	0.26	9	Included	
	Oval	0B	X		E	0.26	9	Included	
Makeup Air Ductwork Exposed in an Unconditioned Space	Rectangular	0B	X		E	0.91	32	Included	
	Round	0B	X		E	0.91	32	Included	
	Oval	0B	X		E	0.91	32	Included	
Supply Air Ductwork Exposed - Outdoors	Rectangular	0B		X	G	0.91	32	Included	5
	Round	0B		X	G	0.91	32	Included	5
Supply Air Ductwork Exposed in Conditioned Space	Rectangular	0B	X		E	0.26	9	Included	
	Round	0B	X		E	0.26	9	Included	
	Oval	0B	X		E	0.26	9	Included	
Supply Air Ductwork (Inside Air temp less than 15 Deg C) Exposed in an Unconditioned Space	Rectangular	0B	X		E	0.91	32	Included	
	Round	0B	X		E	0.91	32	Included	
	Oval	0B	X		E	0.91	32	Included	
Supply Air Ductwork (Inside Air temp above 15 Deg C) Exposed in an Unconditioned Space	Rectangular	0B	X		E	0.71	25	Included	
	Round	0B	X		E	0.71	25	Included	
	Oval	0B	X		E	0.71	25	Included	
Return Air Ductwork Exposed – Outdoors	Rectangular	0B		X	G	0.91	32	Included	5
	Round	0B		X	G	0.91	32	Included	5

FAB Support Buildings and Site Buildings (Note 2)									
Plenum or Ductwork Type	Duct Type	Climate Zone	Plenum or Ductwork Location		Insulation			Jacketing Note 1	Notes
			Indoor	Outdoor	Type(s) Note 1	R-value (min.) $m^2.K/W$	Thickness (basis of design) mm		
Return Air Ductwork Exposed in Conditioned Space	Rectangular	0B	X		E	0.26	9	Included	
	Round	0B	X		E	0.26	9	Included	
	Oval	0B	X		E	0.26	9	Included	
Return Air Ductwork Exposed in an Unconditioned Space	Rectangular	0B	X		E	0.54	19	Included	
	Round	0B	X		E	0.54	19	Included	
	Oval	0B	X		E	0.54	19	Included	
Exhaust Air Ductwork Exposed – Outdoors	Rectangular	0B		X	NR				
	Round	0B		X	NR				
Exhaust Air Ductwork Exposed in an Conditioned Space	Rectangular	0B	X		NR				
	Round	0B	X		NR				
	Oval	0B	X		NR				
Exhaust Air Ductwork Exposed in an Unconditioned Space	Rectangular	0B	X		NR				
	Round	0B	X		NR				
	Oval	0B	X		NR				
Terminal Unit Supply Air Ductwork	Rectangular	0B	X		L2	0.26	9		7
Air Transfer Ducts	Rectangular	0B	X		L2	0.26	9		8
Toilet Room / Janitors Closet Exhaust Ducts in an Unconditioned Space	Rectangular	0B	X		NR				

FAB Support Buildings and Site Buildings (Note 2)									
Plenum or Ductwork Type	**Duct Type**	**Climate Zone**	**Plenum or Ductwork Location**		**Insulation**			**Jacketing Note 1**	**Notes**
			Indoor	**Outdoor**	**Type(s) Note 1**	**R-value (min.) $m^2.K/W$**	**Thickness (basis of design) mm**		
Toilet Room / Janitors Closet Exhaust Ducts in an Conditioned Space	Rectangular	0B	X		E	0.26	9	Included	9
Smoke and CO2 Exhaust Air Ductwork	Rectangular	0B	X		NR				
	Round	0B	X		NR				
	Oval	0B	X		NR				
Staircase, Lobby and Lift well Pressurization Ductwork	Rectangular	0B	X		NR				
	Round	0B	X		NR				

Notes for FAB Support Buildings, Site Buildings and Miscellaneous Applications, Schedule 1 above:

1. Refer to Section 23 07 13 – Ductwork Insulation for insulation and jacketing types, characteristics, and installation.
2. Comply with minimum thermal resistance (R values) as listed in ASHRAE 90.1 (based on climate zone 0B).
3. Maintain vapor barrier over entire blank-off.
4. Provide lining up to 6 meter from the discharge and return air connection of non-clean air handling units. Where internally lined ductwork is shown or otherwise specified, external insulation can be reduced to meet or exceed the overall thermal resistance value of the assembly.
5. Type G insulation includes a white laminate covering (jacket). No field installed jacketing is required. No through-metal attachments allowed. Provide a thermal break to prevent exterior condensation. Maintain the vapor barrier throughout the insulation covering.
6. Only for spaces that are subject to condensation. NR for spaces that are both heated and cooled.
7. Provide lining up to 3 meter from the discharge connection of non-clean VAV terminal units.
8. Provide lining over the entire length of the transfer duct.
9. Provide lining up to 3 meter from each exhaust grille and 3 meter upstream of toilet exhaust fans.

END OF SECTION